Bill of Materials & Component Justification

Project: 1.2m Wingspan RC Aircraft (Conventional High-Wing)
Assignment: Problem 4 Submission

Part A — Identify Components

For a conventional 1.2m high-wing aircraft designed for ground or hand launch, the following systems and components are required to achieve functional flight:

- **Airframe & Structure:** 5mm Depron foam or Sunboard (fuselage/wings), 6mm carbon fiber tube (wing spar), 3mm plywood (motor mount).
- **Propulsion:** Brushless DC (BLDC) outrunner motor, Electronic Speed Controller (ESC), Propeller, LiPo Battery.
- **Control System:** 6-channel 2.4GHz Transmitter (Tx) and Receiver (Rx), four 9g Micro Servos, steel pushrods, clevises.
- **Assembly & Hardware:** Hot melt glue, packing tape, XT60 connectors, 14AWG wire, hook-and-loop straps, rubber bands.
- **Landing Gear:** 3mm piano wire (bent to shape), lightweight foam wheels (2.5 inch), wheel collars.

Part B — Compare Options

Component	Recommendation 1 (Budget)	Recommendation 2 (Performance)	Key Specifications	Qty	Est. Cost (₹)
Motor	Generic A2212 1000KV	Emax XA2212 1400KV	KV rating, max current (A), compatible prop size	1	₹450
ESC	Generic Yellow 30A	Hobbywing Skywalker 40A	Continuous current, 5V/2A BEC	1	₹350
Battery	Orange 3S 2200mAh 30C	Tattu 3S 1500mAh 45C	Cell count (11.1V), capacity, discharge rating	1	₹1,300
Radio (Tx/Rx)	FlySky FS-i6 + iA6B Rx	Radiolink AT9S Pro	6+ channels, 2.4GHz protocol, telemetry	1	₹4,500
Servos	TowerPro SG90 (Nylon)	TowerPro MG90S (Metal)	9g weight, ~1.5kg/cm torque at 4.8V	4	₹400
Propeller	1045 / 10x4.5 Slow Fly	APC 9x4.7 Slow Fly	Diameter and pitch matching motor KV	2	₹120

Part C — Justify Selections

1. **Motor-Propeller Match and Thrust Check:**
   A 1000KV motor paired with a 10x4.5 (1045) propeller on a 3S (11.1V) battery is a proven standard for high-wing trainers. Based on manufacturer data sheets for the A2212 motor series, this combination draws approximately 12–15 Amps at 100% throttle. This setup generates roughly 800g to 900g of static thrust, which is easily sufficient for a 1.2m foam aircraft.

2. **ESC Rating and Electrical Margin:**
   The selected motor draws a maximum of ~15A under heavy load. By selecting a 30A ESC, a 100% electrical safety margin is established. This ensures the ESC will not overheat during sustained full-throttle climbs, especially when enclosed inside a foam fuselage with limited cooling airflow. It also provides reliable power to the receiver via its built-in 5V Battery Eliminator Circuit (BEC).

3. **Battery Selection and Trade-offs:**
   The recommended 3S 2200mAh 30C LiPo prioritizes balance and duration over extreme weight savings. While a 1500mAh battery would be lighter and improve agility, the 2200mAh provides longer flight times (approximately 10–15 minutes). Most importantly, a high-wing trainer naturally tends to be tail-heavy; the 2200mAh battery provides necessary nose weight to properly align the Center of Gravity (CG) without adding "dead weight" like lead ballasts.

4. **Servo Count and Control Surfaces:**
   The design utilizes exactly 4 micro servos. The allocation is: one for the elevator (pitch control), one for the rudder (yaw control), and two for the ailerons (roll control). Utilizing two separate aileron servos (one embedded in each wing half) simplifies mechanical linkages, avoids the binding issues of a single central servo setup, and allows for differential electronic mixing on the transmitter if needed.

5. **Structural Material Selection:**
   5mm Depron or Sunboard (foamboard) is lightweight, highly cost-effective, and easy to manipulate with standard hobby tools. Crucially, foam yields and tears in a crash rather than shattering like balsa wood. This makes it highly repairable on the field using hot melt glue and packing tape. A 6mm carbon fiber spar is added to provide critical lateral stiffness to the 1.2m wing under flight loads.

6. **Compatibility Checks:**
   The FlySky FS-i6 transmitter natively pairs with the iA6B receiver using the AFHDS 2A protocol. The receiver requires an operating voltage of 4.0V–6.5V, which is safely supplied by the 5V/2A BEC integrated into the 30A ESC. Furthermore, the XT60 connectors on the ESC will natively mate with the standard XT60 discharge leads found on the selected 2200mAh 3S LiPo battery, requiring no soldering adapters.

Part D — Budget & Configuration Strategy

1. Total Build Cost Estimation:
Utilizing the "Recommendation 1" (Budget) parts list, the total electrical and radio hardware cost is approximately ₹6,900. Structural materials (foam sheets, carbon spar, glue, pushrods, wire) add roughly ₹600. Therefore, the estimated total build cost is **~₹7,500**.

2. Cost-Saving Opportunities:
The primary cost driver is the radio transmitter (₹4,500). If a transmitter is already available in the lab or shared among a team, the per-airframe cost drops significantly to roughly ₹3,000. Additionally, bending standard piano wire for the landing gear rather than purchasing pre-fabricated carbon fiber gear kits saves money without compromising function.

3. Budget vs. Performance Configurations:

- **Budget Configuration (~₹7,500):** Generic A2212 motor, 30A ESC, nylon gear SG90 servos. Focuses on minimizing financial risk for maiden flights and training.
- **Performance Configuration (~₹9,500+):** Emax motors, Hobbywing ESC, metal-gear MG90S servos. Focuses on longevity, precision control resolution, and durability (metal gears won't strip during rough landings).

4. Final Configuration Choice:
I selected the **Budget Configuration**. Because this is a conventional trainer aircraft likely to be flown by beginners, minor crashes and hard landings are inevitable. Replacing a ₹100 nylon servo or a generic motor after a nose-in crash is much more economically sustainable. The aerodynamic envelope of a slow-flying high-wing plane does not require the micro-precision of premium electronics.

> ### Bonus (+5): Mass Estimation & Propulsion Assessment
>
> **Mass Estimation:** Airframe Foam (~250g) + Battery (~185g) + Motor (~50g) + ESC (~30g) + Receiver/Servos (~50g) + Hardware/Landing Gear (~85g) = **Estimated All-Up Weight (AUW) of ~650g.**
>
> **Propulsion Assessment:** The A2212 1000KV motor with a 1045 prop generates ~800g of thrust. This results in a thrust-to-weight ratio of approximately **1.23 : 1**. Since a standard trainer requires only a 0.7:1 ratio for level flight, a 1.23:1 ratio provides ample excess power. This proves the propulsion is more than sufficient for rapid altitude recovery and easy hand-launches.